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COMMENTARY

Core GRADE unpacked: a summary of recent innovations in complementary GRADE methodology

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Abstract

The Grading of Recommendations Assessment, Development and Evaluation (GRADE) framework has become the global standard for rating evidence certainty and grading strength of health-care recommendations in systematic reviews, clinical practice guidelines (CPGs), and health technology assessments (HTAs). However, as the methodology has evolved, its growing complexity and difficulties navigating the many articles providing GRADE guidance have created challenges for users. Moreover, GRADE guidance has been published across multiple journals and platforms, resulting in a body of work with considerable redundancy, modifications, and out-of-date material that requires meticulous appraisal of GRADE writings to identify and implement current guidance. In response, a group of GRADE methodologists developed Core GRADE, a simple yet comprehensive framework focused on the essential elements required to apply GRADE to paired comparisons of interventions. Building on prior GRADE publications, the authors reviewed existing guidance and distilled its fundamental components. This proposal represents an independent approach and has not received formal endorsement from the GRADE working group. During this process, they identified multiple areas where clearer recommendations were warranted and incorporated these improvements into the seven Core GRADE papers. This article presents the resulting innovations introduced by core GRADE, that include the use of flow charts and algorithms to guide GRADE implementation; an emphasis on viewing both individual GRADE domains and overall certainty as continua; and clarification of decisions related to addressing potential relative and absolute subgroup effects when formulating patient population, intervention, comparison, and outcome questions. © 2025 Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Keywords: GRADE; Core GRADE; Systematic reviews; Clinical practice guidelines; Health technology assessments

1. Introduction

The Grading of Recommendations Assessment, Development and Evaluation (GRADE) framework has become the global standard for rating evidence certainty and grading strength of health-care recommendations in systematic reviews, clinical practice guidelines (CPGs), and health technology assessments (HTAs) [1].

GRADE is a living system of guidance that has undergone, and continues to undergo, constant revision. The

Core GRADE comes not as an official product of the GRADE Working Group, but rather as the collaborative effort of a group of GRADE experts, building on their pivotal roles in much of previous GRADE guidance, in applying GRADE through the MAGIC Evidence Ecosystem Foundation, in developing BMJ Rapid Recommendations, and in innumerable systematic reviews and guidelines.

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Plain Language Summary

The GRADE framework was introduced in 2004 to help researchers and health-care professionals assess the certainty (quality) of evidence in systematic reviews and the strength of recommendations in CPGs and HTAs. Over the past 2 decades, it has become overwhelmingly the most widely used approach to making certainty of evidence and strength of recommendation decisions. However, many users now find GRADE increasingly complex, and guidance has appeared in numerous journals and platforms. This has created overlap, outdated information, and inconsistencies that make it difficult to identify the most current best approaches. To address these challenges, GRADE experts developed Core GRADE, a simplified but comprehensive version of GRADE. Core GRADE brings together the essential elements needed to apply GRADE when comparing health-care interventions (1 intervention vs 1 comparator). In addition to this simplification, Core GRADE offers several innovations. These include practical flowcharts and algorithms to guide step-by-step application of the framework; a new emphasis on viewing both individual domains and overall certainty of evidence as continua; and clearer advice on how to address different subpopulations when formulating research questions. Beyond simplification, these innovations further enhance GRADE's usefulness and ease of use.

essence of GRADE guidance appeared in the first 15 articles in the JCE GRADE series [2]. Subsequently, important modifications have included the introduction of guidance for diagnostic tests [3,4], for value and preference studies [5,6], use of ROBINS-I [7], handling of missing data [8], evaluation of cost-effectiveness [9], evaluation of network meta-analysis [10−13], time to event analysis [14], updated guidance regarding rating down for inconsistency [15], and most fundamentally and dramatically, change in the target of certainty rating when applying GRADE [16] and subsequent modification in guidance regarding rating down for imprecision [17,18]. This is not a complete list, and a considerable portion of this updated guidance has proved sufficiently complex that it is seldom applied by GRADE users.

In response, a group of leading GRADE methodologists created Core GRADE, a concise, structured, and user-friendly approach which streamlines GRADE's application when addressing the impact of a single intervention against a comparator. A series of 7 papers [19−25] published between April and June 2025 in the BMJ provide a comprehensive presentation of Core GRADE [26]. Core GRADE represents the latest evolution in take it or leave it living GRADE guidance, but differs fundamentally from previous updates in presenting a complete one-stop-shopping simplified approach to the use of GRADE that emphasizes clarity and usability. Core GRADE is not an official product of the GRADE working group, but represents a collaborative effort of GRADE experts, drawing on their contributions to prior GRADE guidance. The series also includes a number of implementations tools beyond what GRADE has previously provided. This paper summarizes those innovations.

1.1. Algorithmic structuring

A key innovation in Core GRADE is the introduction of detailed flowcharts and algorithms that offer a visual roadmap to guide users step-by-step through the GRADE process. These implementations tools clarify procedures, reduce ambiguity, and make the use of GRADE more systematic and accessible. They also exposed areas of vagueness in previous guidance, and addressing those limitations has achieved greater explicitness and clarified GRADE guidance.

The modification of prior guidance in the context of an algorithm is best captured in presenting Core GRADE's approach to rating risk of bias in a body of evidence [22]. For many years, systematic reviewers faced challenges in deciding whether or not to rate down certainty of evidence due to risk of bias. In the absence of a clearly articulated GRADE approach, authors sometimes used varying and potentially problematic approaches [22]. Core GRADE represents the first time that GRADE offers a clear algorithm to, after classifying each study as low or high risk of bias, arrive at the decision whether to rate down across the body of evidence.

The algorithm (Fig 1) suggests first deciding if the evidence is dominated by high risk of bias studies—the explicit suggestion for ''dominated'' being more than 65%, with 55% as an alternative, of the weight in the meta-analysis comes from high risk of bias studies. The thresholds were developed based on real examples and discussions within the Core GRADE group, with the intention of providing flexible guidance to explore the relationship between the study risk of bias and its potential impact on the direction and interpretation of the effect estimate [22].

If the answer is yes, with one exception, authors need to pool results from all studies as the best estimate and rate down the certainty of evidence for risk of bias. The exception is as follows: (i) the pooled estimate shows an effect and, considering the direction of bias, the real effect is the same or larger than the estimated effect or (ii) the result shows no effect and, considering the direction of bias, the real effect is the same or even smaller. Under each of these circumstances, rating down is not required. The Core

What is new?

Key findings

- In addition to its fundamental goal of simplifying Grading of Recommendations Assessment, Development and Evaluation (GRADE), the seven-part British Medical Journal Core GRADE series includes a substantial number of important innovations in the GRADE framework for rating certainty of evidence and moving from evidence to recommendations.

What this add to what is known?

- Most users of the Core GRADE will not carry a detailed knowledge of prior GRADE guidance. As a result, they will not be aware of what is only a simplification of prior guidance and what is innovative. In reflecting or referring to prior GRADE writings, it will be useful for Core GRADE users to be aware of what is new and innovative in the Core. These innovations include the use of step-by-step algorithms, viewing certainty of evidence as a continuum rather than a fixed points, clearer guidance on subgroup issues when framing patient population, intervention, comparator, and outcome questions, ways to deal with both absolute and relative effects, strategies for situations with low certainty of evidence, a simpler approach to judging imprecision, and simpler steps for moving from evidence to recommendations highlighting the importance of the minimal important difference.

What is the implication and what should change now?

- Core GRADE offers a streamlined alternative to the full GRADE methodology, and awareness of what is new will help Core GRADE users to make the optimal use of the great potential usefulness of the Core GRADE series.

GRADE risk of bias paper provides worked examples of these situations [22]. When appreciable evidence exists from low risk of bias studies (more than 35% or 45% of the weight in the meta-analysis comes from low risk of bias studies), review authors need not rate down for risk of bias. In this circumstance, however, the question remains as to whether systematic reviewers should use results from all studies or from only low risk of bias studies.

To address this issue, Core GRADE suggests testing whether results are similar vs. differing importantly for the low and high risk of bias studies. If there are important differences, review authors will use only low risk of bias studies and not rate down for risk of bias. If similar, they will use all studies, and once again not rate down for risk of bias [22]. The algorithm (Fig 1) captures each of these steps.

Across the completed series, Core GRADE presents a set of algorithms designed to streamline the evaluation of evidence and its translation into clinical recommendations. In addition to the risk of bias algorithm, imprecision provides two algorithms: the steps for selecting the threshold and target of interest and the steps for rating imprecision [20]. The inconsistency flow chart presents the roles of point estimate variability and overlapping CIs as well as the location of the point estimates in relation to the threshold chosen for rating certainty [21]. Finally, the last article provides an algorithm for moving from evidence to recommendations, highlighting the role of establishing minimal important differences for each outcome [25].

1.2. Certainty of evidence as a continuum

A key contribution in the series emphasizes that concepts of evidence trustworthiness, although presented as categories, in fact represent underlying continua [19]. This is true for ratings of individual domains (imprecision, inconsistency, risk of bias and indirectness), with its categories of no serious, serious, very serious, and extremely serious limitations (Fig 2A). The continuity concept also holds true for ratings of overall certainty of a body of evidence addressing a particular outcome (eg, mortality, quality of life) (Fig 2B). These figures highlight how decisions regarding rating down an individual domain, and rating overall certainty as high, moderate, low, or very low, often represent close calls.

Although briefly acknowledged in earlier GRADE publications, the introductory article in Core GRADE places considerable emphasis on how close-call decisions can have profound implications for overall ratings [19]. For instance, one may have close-call decisions on three domains and, looking individually at each, choose not to rate down in any. One would therefore finish with high-certainty evidence. Alternatively, one may choose in these three close-call decisions to rate down for all three domains and finish with very low-certainty evidence. Both would be problematic. Viewing each domain as a continuum and taking a gestalt or overall view of the evidence, reviewers would likely conclude that the evidence is near the threshold between moderate- and low-certainty evidence. They could then choose one of these two options, noting that the rating is near the threshold between the two categories. The Core GRADE overview paper provides an example of such a situation [19].

Core GRADE adds an explicit recognition that decisions about whether to rate down for individual domains are often "close calls," and that treating these judgments as existing on a continuum helps reviewers avoid extremes (eg, moving from high to very low certainty due to multiple

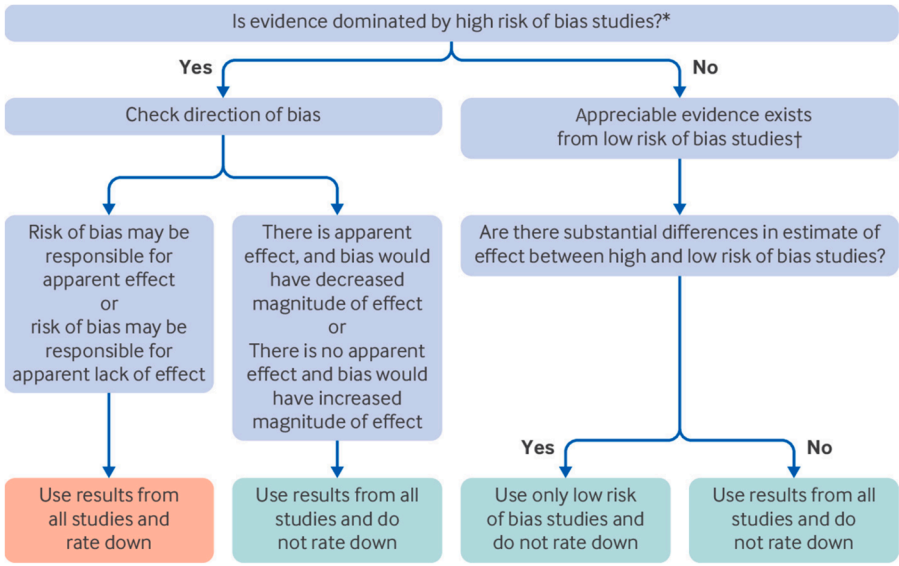

Figure 1. Algorithm for determining whether to rate down for risk of bias. *Possible thresholds for high risk of bias "dominating": >65% weight or ≥55% weight = possibly dominating. †Possible thresholds for appreciable low risk of bias evidence, corresponding to thresholds for high risk of bias dominating, ≥35% or ≥45% weight.

borderline decisions). By adopting a gestalt view of the evidence, reviewers can place their judgment closer to the threshold between categories (g, between moderate and low), and explicitly acknowledge when a rating represents such a borderline case. This approach preserves the practicality of categories while recognizing the underlying continuum on which certainty of evidence truly lies.

1.3. Addressing possible subgroup issues when formulating the intervention, comparator, and outcome questions

In framing the patient population, intervention, comparator, and outcome (PICO) question, systematic reviews, CPGs, and HTA authors must consider the possibility that relative effects differ across patient subgroups (eg, effects may differ in old vs. young people) or across interventions subgroups (eg, oral vs. parenteral antibiotic treatment). To address the issue, Core GRADE for the first time clarifies three possible scenarios: (i) one has no reason to suspect differences in effects across subgroups; (ii) one is confident that effects vary across subgroups; or (iii) one has good reason to suspect subgroup differences but is uncertain.

Core GRADE exemplifies these three possibilities in Table 1 [21]. This clarification is likely to be helpful for authors as they frame their PICO question.

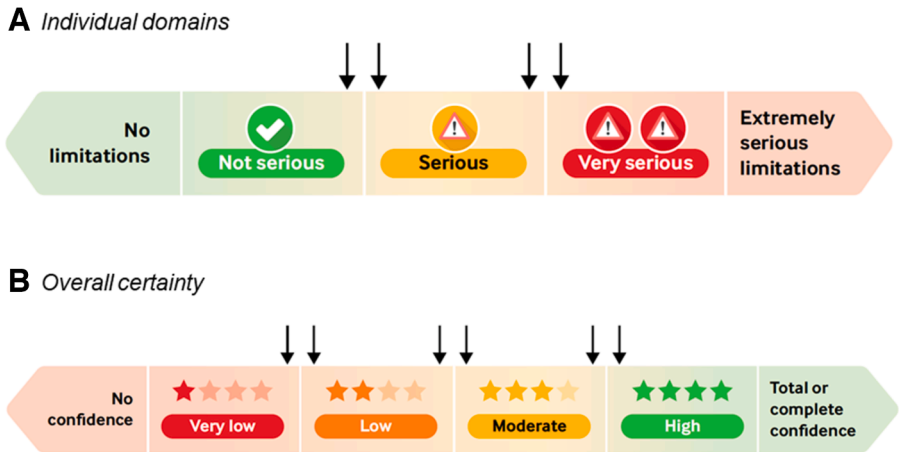

Figure 2. Certainty of evidence as a continuum. (A) Categories for rating down individual domains reflect an underlying continuum. (B) Categories for rating certainty of evidence down reflect an underlying continuum.

Table 1. Three scenarios when considering subgroups during PICO construction

Scenario	Implications for PICO construction	Example
1. Previous research provides no compelling evidence that effects should be different across patient or intervention subgroups (no subgroup hypothesis)	Combine all subgroups (single estimate of effect) without a subgroup hypothesis	The World Health Organization has generated several recommendations regarding the management of patients with COVID-19. The guideline panels inferred that effects were very likely to be similar in men and women and thus in all their recommendations provided a single estimate for men and women [27].
2. Previous research suggests that effects differ across patient or intervention subgroups (subgroup effects are presumed to exist)	Narrow PICO to one subgroup, or construct two separate PICOs for each subgroup	A guideline panel addressing optimal transfusion thresholds in anemic patients considered that the biology differed between children and adults and therefore looked at the evidence separately and provided separate recommendations [28].
3. Previous research plausibly suggests that effects differ across patient or intervention subgroups, but one is uncertain (directional subgroup hypothesis)	Initially combine all subgroups (single estimate of effect), but also provide and then test a directional subgroup hypothesis	A systematic review comparing immediate vs. delayed antiretroviral therapy in patients with a concomitant diagnosis of HIV and tuberculosis tested whether the impact of early vs. delayed treatment on mortality differed between those with higher and lower CD4 cell counts. A previous trial suggested that hypothesis, including a clear direction, but for another outcome [29].

PICO, patient population, intervention, comparison, and outcome.

1.4. Specifying hypotheses regarding both relative and absolute effects as part of the PICO development process

Core GRADE recommends that systematic review authors and guideline panels clearly distinguish between relative and absolute effects, including when exploring the possibility that treatment effects could differ across patient, intervention, or outcome characteristics. That is, differences in *relative effects* (eg, a relative risk [RR] of 1.0 in one group and 0.5 in another) vs. *absolute effects* derived from variations in *baseline risk* (eg, RR remains constant but the baseline risk of the outcome is 5% in one group and 20% in another).

For differences in relative effects, to avoid undermining credibility, review teams should limit the number of hypothesized subgroup effects (ideally $\leq$3), specify their direction, and recognize that true differences in relative effects are uncommon. They should avoid excessive and nondirectional hypotheses about subgroup effects.

In contrast, absolute effects frequently vary across patient groups due to differences in baseline risk of the outcome and absolute effects are those most relevant for patient care decisions. For example, a consistent 50% RR reduction may translate into a 1% absolute benefit in low-risk patients and a 20% benefit in high-risk patients, leading to very different clinical decisions.

For instance, the WHO (World Health Organization) COVID-19 guideline anticipated that both relative and absolute effects for nirmatrelvir-ritonavir will be different for particular patients' characteristics (subgroups) and explicitly hypothesized that relative benefit would decrease with delayed administration (Table 2).

The systematic review team reported no credible relative subgroup effects [31]. They did, however, identify substantial differences in effect in patients with different baseline risks. In particular, they found that despite having the same relative effect, for patients at high baseline risk nirmatrelvir-ritonavir is likely to affect important reductions in both mortality and hospitalizations; in low-risk patients, benefits are likely to be trivial. More specifically, the anticipated reduction in hospitalization for high risk was 50 per 1000 (5%) and for low-risk patients approximately 4 per 1000 (0.4%), and when considering the results demonstrating the large differences in the effect between high- and low-risk patients, the guideline panel issued a strong recommendation in favor of nirmatrelvir-ritonavir for high-risk patients and a weak recommendation against its use for low-risk patients.

Table 2. Illustrative PICO question incorporating hypotheses about both relative and absolute treatment effects

WHO Covid-19 CPG [30]: Nirmatrelvir-ritonavir PICO question	
Population	Patients with nonsevere confirmed COVID-19 (according to WHO severity definitions)
Intervention	Nirmatrelvir-ritonavir + standard care
Comparator	Standard care
Outcomes	Hospitalization, mortality, mechanical ventilation, adverse effects leading to drug discontinuation, time to symptom resolution
Groups with possible varying baseline risk and thus different absolute effects (prognostic factors expected to lead to differences in baseline risk)	Likely determinants of baseline risk: 1. Larger baseline risk and therefore larger beneficial effects of the intervention in the immunosuppressed or those with chronic disease 2. Larger baseline risk in those with negative serology 3. Larger baseline risk in the unvaccinated
Groups with possible varying relative risk (speculative differences in effect in different patient groups)	Candidate effect modifiers: 1. Larger beneficial relative effects of the intervention in those with shorter time from symptom onset 2. Larger beneficial effects in those with negative serology 3. Larger beneficial effects in the unvaccinated

PICO, patient population, intervention, comparison, and outcome; WHO, World Health Organization.

1.5. Addressing situations with expected low and very low certainty of direct evidence

Core GRADE offers guidance on how guideline developers should prepare in advance when they anticipate finding only low or very low certainty direct evidence [23]. Under these circumstances they should formulate specific PICOs for indirect evidence that may ultimately prove the highest-certainty evidence available. For instance, early in the COVID-19 pandemic before randomized controlled trial (RCT) evidence was available, guideline developers addressing corticosteroid administration in those with severe COVID-19 conducted a formal search for evidence from RCTs in patients with acute respiratory distress syndrome (ARDS), a clinical presentation similar to critically ill patients with COVID-19 [32]. Although limited by indirectness, the panel judged the high-certainty evidence of benefit from corticosteroid in the ARDS patients was sufficient to issue a conditional recommendation for the use in patients with severe COVID-19 [33].

In a related issue, Core GRADE authors noted that the evidence emerging from a targeted search for indirect evidence will almost invariably result in rating down for indirectness. In contrast, the degree of indirectness in PICO one finds in studies identified through a search for direct evidence will seldom warrant rating down certainty of evidence. Finally, Core GRADE highlights that although benefits are likely to vary across conditions, harms are, in contrast, likely to be similar [23]. Thus, pooled adverse event data when the intervention of interest is administered in RCTs of related conditions represent a particularly valuable source of indirect evidence [34–37].

1.6. Clarification of inconsistency vs indirectness

Core GRADE clearly distinguishes between addressing inconsistency and addressing indirectness. Inconsistency is defined as unexplained variability in results across studies in the meta-analysis. In addressing inconsistency, Core GRADE users examine whether observed differences reflect true heterogeneity or random error and, having ruled out, random error undertake an evaluation of relative subgroup effects and a judgment of their credibility (eg, evidence is available for both older and younger patients) [21]. In contrast, indirectness arises when there is a mismatch between the target PICO and the available evidence. It evaluates the extent to which such discrepancies compromise the applicability of the evidence to the clinical or policy decision-making context (g. one is interested in the old, but the only available evidence is in the young) [23].

1.7. Clarification in selective outcome reporting

Core GRADE offers important clarification in the evaluation of selective outcome reporting, distinguishing between two distinct forms of bias [22]. The first involves outcomes that investigators measured but did not report, an issue representing a subtype of publication bias. The second involves outcomes that were reported but selected from multiple possible effect estimates, such as different time points, outcome measurement methods, or analytical approaches. In some cases, discrepancies may also arise between what was specified in the study protocol and what appears in the final publication [22]. This represents selective reporting in individual studies (ie, risk of bias), but not publication bias. This clarification allows GRADE users to accurately classify limitation as risk of bias vs. publication bias.

1.8. The importance of the minimally important difference in moving from evidence to recommendations

In Core GRADE, the use of MIDs becomes a cornerstone in the transition from evidence assessment to

recommendation formulation. Core GRADE users should specify MIDs for all outcomes a priori. Core GRADE suggests a practical method for identifying MIDs: the panel survey approach, which engages guideline developers and content experts in systematically rating what constitutes an important difference [25]. This method helps ensure that the chosen MIDs are transparent, reproducible, and aligned with patient values and preferences.

1.9. Simplification of the evidence to decision framework

By clearly identifying which elements require attention depending on the perspective, Core GRADE supports more context-sensitive and efficient decision-making [25]. Core GRADE specifies that users often have no need to complete the full evidence-to-decision (EtD) checklist, but rather can provide narratives for each criterion [25]. Moreover, they can restrict this commentary to key domains. For individual patient care, consideration of benefits and harms, certainty of evidence, and patients' values and preferences may be sufficient to inform recommendations. In contrast, for public health and health system perspectives, panels must also consider additional EtD domains, including resource use, feasibility, acceptability, and equity [25].

1.10. Methodological simplifications

Core GRADE, although maintaining alignment with foundational GRADE principles, provides methodological simplifications to improve usability. Core GRADE specifies that nonrandomized studies of interventions should initially be rated as low-certainty evidence, taking a simpler approach than frameworks such as ROBINS-I that allow for high initial certainty with subsequent rating down [22]. Core GRADE also supports the use of the newer, more streamlined ROBUST-RCT tool for assessing risk of bias of RCTs, preferring its simplicity over the more complex Cochrane Risk of Bias 2 tool [22]. Finally, Core GRADE includes only two of the three criteria for rating up the certainty of evidence (large magnitude of effect and dose-response gradient), and omits the third that pertains to the effect of plausible residual confounding, noting that it is very seldom invoked and, when it is, may be at best controversial [19].

Although any simplification facilitates implementation, it also carries the risk of being less accurate. Core GRADE does carry this risk. However, Core GRADE does not emerge as a new approach without prior experience. The authors of the Core GRADE papers have been aware for a considerable time that GRADE has become too complicated and, in their GRADE practice, have implemented the principles that ultimately emerged as the BMJ's Core GRADE series. This implementation has played out in work with many guideline groups, including physicians' organizations and other such as UpToDate, and has proved extremely successful in resulting in GRADE application that consistently proved of top quality. Indeed, it is this success that motivated us to produce the Core GRADE series. Thus, evidence suggests that the risk of reduced accuracy with use of the Core is minimal.

2. Conclusions

Core GRADE marks a significant advance in the evolution of the GRADE framework, delivering a streamlined yet comprehensive approach to assessing the certainty of evidence and the strength of recommendations. Core GRADE introduces a number of important innovations that strengthen an updated GRADE methodology and facilitate its wider adoption and effective use by guideline developers, systematic reviewers, and decision-makers across a variety of health-care settings.

CRediT authorship contribution statement

Luis Enrique Colunga-Lozano: Writing − review & editing, Writing − original draft, Project administration. **Ying Wang:** Writing − review & editing, Validation, Conceptualization. **Thomas Agoritsas:** Writing − review & editing, Validation, Conceptualization. **Monica Hultcrantz:** Writing − review & editing, Validation, Conceptualization. **Alfonso Iorio:** Writing − review & editing, Validation, Conceptualization. **Victor M. Montori:** Writing − review & editing, Validation, Conceptualization. **Gordon Guyatt:** Writing − review & editing, Writing − original draft, Supervision, Methodology, Conceptualization.

Declaration of competing interest

There are no competing interests for any author.

Data availability

No data was used for the research described in the article.

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